Ashish Behera

Mechanical Engineering

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Professional Summary

Mechanical Engineering graduate with hands-on experience in inspection and quality at Hindustan Aeronautics Limited (HAL). Worked on KRD-99B engine health assessment, engine sensor assembly, AL-31FP engine system testing, and hydraulic system assembly and testing. Interested in mechanical design, aerospace quality, industrial systems, and practical automation. My academic work includes a motorized scissor jack with load and safety monitoring and a forklift concept focused on safe navigation, object detection, and warehouse operations.

Professional Experience

Hindustan Aeronautics Limited (HAL) Koraput, Odisha
Apprentice – Junior Engineer *July 2025 – July 2026*

- Assisted in KRD-99B engine health-related inspection work and engine sensor assembly tasks.
- Supported AL-31FP engine system testing and related quality inspection activities.
- Contributed to hydraulic system assembly and testing as part of aerospace engineering work.
- Gained practical exposure to component handling, assembly procedures, system testing, and quality control practices in an industrial aerospace environment.

Academic Projects

Smart Motorized Scissor Jack with Safety Monitoring
Parala Maharaja Engineering College

- Designed a motorized scissor jack with an electric lifting mechanism and load-awareness features.
- Added safety alerts to warn the user when the system approached unsafe operating conditions or potential failure.
- Included a buzzer-based alert system to indicate overload, motor stress, or unusual operating conditions.
- Proposed a basic web interface for fault indication and system monitoring to improve usability and maintenance.
- Focused on safe lifting, load awareness, fault notification, and practical monitoring for real-world use.

Autonomous Forklift Concept
Uma Charan Patnaik Engineering School

- Developed a concept for an autonomous forklift intended for material handling and warehouse transportation.
- Explored object detection, navigation, and image-based recognition to improve awareness of the surrounding environment.
- Included path detection and shortest-route planning concepts to support efficient movement between storage points.
- Proposed a web-based control interface for warehouse operations, movement control, and monitoring.
- Incorporated battery status and payload monitoring to support safe and efficient operation.
- Included emergency-stop logic and braking concepts to improve overall safety and reduce collision risk.

Technical Skills

CAD and Mechanical Design: SolidWorks, AutoCAD, CATIA

Engineering Analysis: ANSYS, Engineering Mechanics, Tolerance Analysis

Aerospace and Quality: Component Inspection, Engine Sensor Assembly, System Testing, Quality Control

Mechanical Systems: Hydraulic Systems, Lifting Mechanisms, Load Monitoring

Automation and Control: Image Processing, Object Detection, Path Planning, Autonomous Material Handling

Monitoring and Interface: Battery Monitoring, Load Monitoring, Braking Concepts, Web-Based Control

Software and Productivity: IFS ERP System, Microsoft Office, LibreOffice

Professional Skills: Teamwork, Leadership, Problem-Solving, Time Management

Education

Parala Maharaja Engineering College — Berhampur, Odisha, India
B.Tech. in Mechanical Engineering; GPA: 7.35 — *November 2021 – June 2024*

Uma Charan Patnaik Engineering School — Berhampur, Odisha, India
Diploma in Mechanical Engineering; Percentage: 71.60% — *July 2018 – August 2021*

Certifications

The Complete C Programming Course for Beginners: By Md. A. Barik | Udemy.com 2022

Java Complete Course For Begineers 2022: Global Code Institute | Udemy.com 2022

C++ Programming Step By Step From Beginner To Ultimate Level Instructor: Lernen Hub Academy | Udemy.com 2022

Python Programming Complete Beginners Course Bootcamp: By Dmitry Yedunov | Udemy.com 2022

Hobbies & Interests

Programming: Exploring Software Development, Coding, and Problem-Solving

Embedded Systems: Building Embedded Projects and Exploring Microcontrollers

Electronics Projects: Designing and Experimenting with Electronic Circuits and Hardware

Android Development: Exploring Android App Development and Mobile Technologies